

Introduction to blockchain technology

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1.1 Introduction

Blockchain was first introduced by Nakamoto (2017) to serve as a cryptocurrency exchange record (Liu et al., 2019). This is a decentralized, dispersed, and secure record without the need for a trusted central reference. The distributed database contains a regular list of blocks that are connected through links called chains (Gorkhali et al., 2020). This chain structure ensures that the entire transaction history on a blockchain is tamper-proof. Each block contains a hash of the previous block, making it nearly impossible to tamper with past transactions (Goncziol et al., 2020). The very nature of how blockchains are built (architecture and design) inherently offers security benefits (Casino et al., 2019). These features have led to the use of this innovation in different areas such as fiscal (Treleven et al., 2017; Vandezande, 2017), healthcare (Mettler, 2016; Zhao et al., 2017), cloud computing (Sharma et al., 2017), intelligent transportation systems (Kang et al., 2017), power grid (Peck & Wagman, 2017; Rottondi & Verticale, 2017), and supply chain (Nakasumi, 2017).

Blockchain-based systems can arrange interactions of commonly doubting parties without a trusted third party (Wagner et al., 2019). Blockchain has the potential to be a game-changer for the future of cybersecurity by serving as the foundation for a more secure digital landscape (Qasse et al., 2019). This technology provides a consensus, provably secure, and distributed solution to solve traditional vulnerabilities. Therefore it can provide security services such as privacy and authentication. In addition, it can prepare great security for extensive communications and transactions (Nguyen, n.d.). At present, these tasks are presented by trusted third parties or utilizing dispersed approaches (Bhushan et al., 2021). Fig. 1.1A presents a centralized approach that uses trusted third-party and Fig. 1.1B illustrates the blockchain-based approach which uses consensus algorithms.

Currently, many blockchains perform in an autonomous and disconnected environment with each other (Qasse et al., 2019); thus interoperability between blockchains becomes a critical challenge. There exist several methods to handle the issue of the cooperative

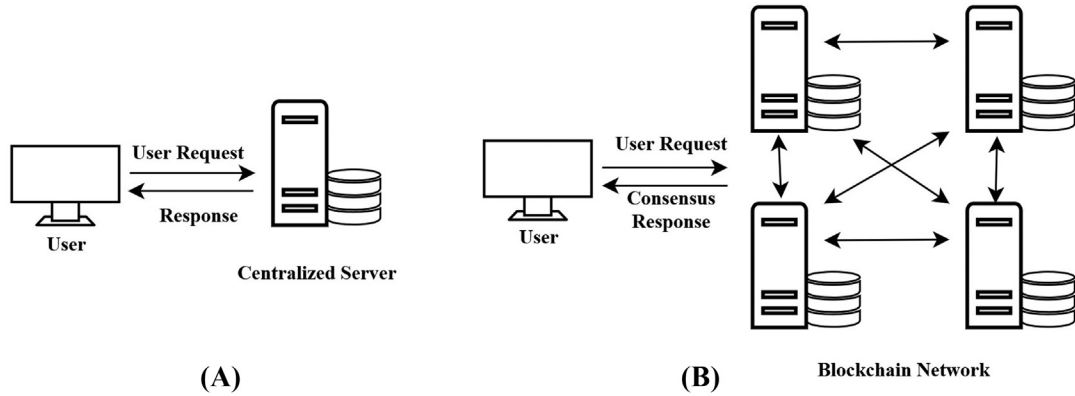


FIGURE 1.1 Access control systems: (A) centralized and (B) blockchain-based.

operation between distinctive blockchains, such as (Wang et al., 2022) virtual chain, which is proposed to support the cross-chain transaction, and also an approach that is presented by Jiang et al. (2019) to combine numerous blockchains for managing IoT data. Establishing trustless cross-blockchain trading protocols is challenging. As employing centralized exchanges to transfer across blockchains needs trust, this undermined the nature of the blockchains on which they operate. Therefore a trustless decentralized exchange is required to support the transfer of cross-blockchains (Zamyatin et al., 2019). Various strategies have been presented to guarantee the security of the blockchain. There exist some overviews on these techniques, such as Hathaliya and Tanwar (2020), Ma et al. (2020), and Zheng et al. (2019); additionally, there are several surveys on the cross-blockchain approaches such as Borkowski et al. (2019), Qasse et al. (2019), and Vo et al. (2018). In this chapter, blockchain, cross-blockchain, security challenges, and threats in the blockchain are reviewed.

The remaining sections of the chapter are arranged in a systematic manner and are divided into five sections for better organization, as expressed in Fig. 1.2. Section 1.2 dives into the fundamentals of blockchain technology, exploring its various types and the concept of cross-blockchain interaction. Section 1.3 illustrates potential applications. Section 1.4 discusses security services and the suggested blockchain solutions. Section 1.5 explains the blockchain challenges and security threats on the blockchains. Eventually, in Section 1.6, we terminate this study.

1.2 Blockchain technology

This section aims to explain the fundamental concepts and important terminologies related to blockchain technology. It is aimed at familiarizing the reader with the essential terminologies and ideas related to the topic. A brief introduction to blockchain technology consists of a definition, an explanation of some important elements, and initial presentation of blockchain process. Following that, the types of blockchains and cross-blockchains are explained.

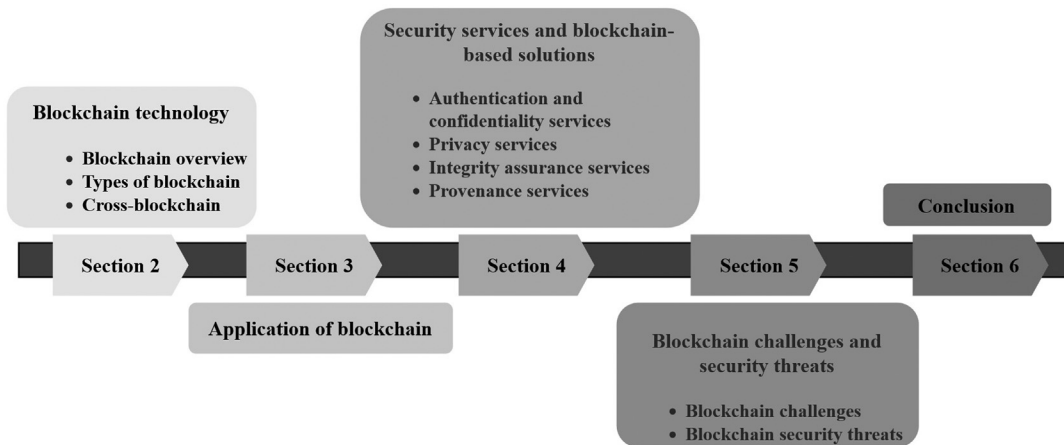


FIGURE 1.2 Overview of the chapter.

1.2.1 Blockchain overview

The shared ledger concept is a core principle that empowers blockchain technology to function as a secure and transparent platform for recording and tracking information. This innovation permits us to record data for all time and without alteration. Some of the considerable components of this technology are as follows (Bernabe et al., 2019):

Transaction: In a blockchain, a transaction is a record added to the ledger that captures information about an operation.

Block: A block in a blockchain acts like a container, storing several elements: transaction data, a timestamp indicating when the block was created, cryptographic hashes of the preceding block, and additional information like digital signatures and a nonce value.

Timestamp: The creation or modification time of a document can be tracked using the required timestamp, which is included in the block created.

Hash: The hash function encrypts input and generates a unique fixed-length value.

Genesis block: The starting point of a blockchain, distinguished by the absence of a prior block hash.

Chain: The chain is the linked list of blocks that starts with the genesis block.

Network: A decentralized network of devices, called nodes that directly communicate with each other.

Consensus: To ensure everyone agrees on the state of the blockchain, nodes in the network rely on a consensus algorithm. This mechanism allows them to collaborate and reach a common understanding on the validity of transactions and the order they appear in the ledger.

Immutability: It means that once the information is written in a block nobody can modify it.

In Fig. 1.3, the blockchain process is explained with an example. When conducting a blockchain transaction between User A and User B, a new block must first be created containing the transaction details. A completed block is broadcast to the network for

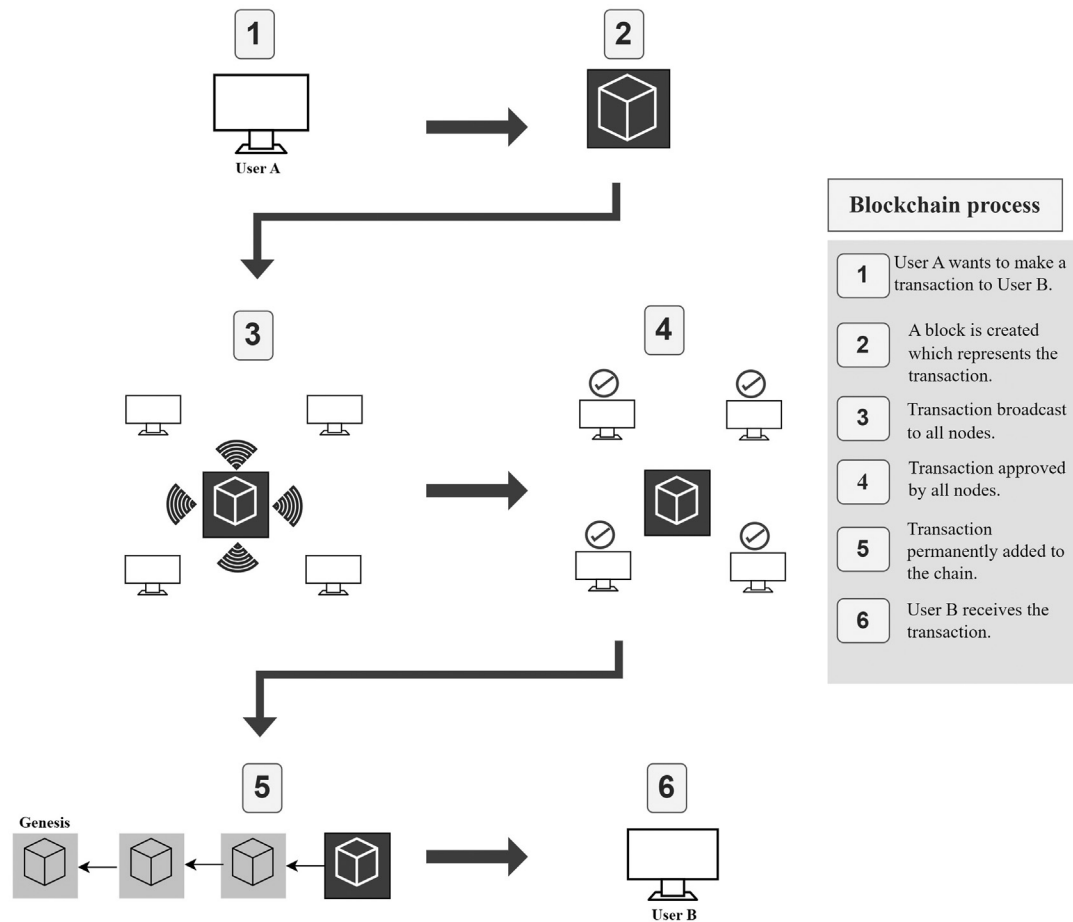


FIGURE 1.3 Blockchain process (Rayenizadeh et al., 2021).

validation. Once confirmed through the consensus mechanism, it becomes a permanent part of the blockchain and is distributed to all participants. As a result, User B receives the transaction, and other nodes update their blockchain accordingly. Transactions that are not validated by the consensus mechanism will not be included in the blockchain and are eventually discarded (Zubaydi et al., 2019).

1.2.2 Types of blockchain

Blockchains are classified into three main groups according to their different applications (Zheng et al., 2019), which are explained below.

A Public blockchain is open and relies on consensus mechanisms for governance. Anyone with an internet connection can join the network as a node and participate in

transactions. Transparency and anonymity are the characteristics of transactions and every participant can read them.

A Private blockchain is centralized and used in an organization to process internal transactions. Only organizations have writing permissions, and only the owner retains the writing permission, and no agreement is required. But reading permissions are granted to a limited group of participants or it can be public.

Federated blockchain, sometimes called consortium blockchain, operates under the governance of a predefined group of trusted organizations. Similar to permissioned blockchains, only these authorized members have the right to add data (write) to the ledger. The consortium collaboratively validates transactions. Read access can be public or restricted depending on the specific participants.

1.2.3 Cross-blockchain

Connecting several blockchain networks without any restrictions and transferring information between them is called interoperability in a blockchain network (Qasse et al., 2019). Recently, solutions have been proposed for the interconnection of blockchains (Wang et al., 2020). Enabling data and asset transfer between different blockchains can revolutionize blockchain technology (Ghaemi et al., 2021). Cross-blockchain transaction (Wang et al., 2020) is a distributed protocol for conducting transactions in the optional number of blockchains and has no centralized component. Sigwart et al. (2021) presented a cross-blockchain protocol to transfer the asset. The proposed structure in Nissl et al. (2021) invokes smart contracts across blockchains. Qiu et al. (2019) has proposed a method (ChainIDE) for improvement of smart contracts based on blockchains on different types of blockchain systems.

1.3 Applications of blockchain

The potential applications of blockchain technology are constantly evolving. As the technology matures and gains wider adoption, we can expect to see even more innovative uses emerge across various industries. In the following, we review some applications of the blockchains:

- Financial: Blockchain technology has found many applications today, and Bitcoin is the first to use (Nakamoto, 2017). This technology is used in different financial fields such as business services and economic transactions (Haferkorn & Quintana Diaz, 2015). It can be a promising technology for improving and developing a sustainable global economy (Casino et al., 2019; Nguyen, 2016).
- Healthcare: Security and interoperability in healthcare are essential, and blockchain prepares strong solutions for these systems (Soltanisehat et al., 2020). A particularly exciting area of exploration for researchers is the potential application of blockchain technology within the healthcare industry (Boulos et al., 2018; Hölbl et al., 2018). In addition, several researchers have reviewed the proposed methods and explored the challenges and opportunities in this area such as Houtan et al. (2020), Omar et al. (2021), Shi et al., 2020, and Tandon et al. (2020).

- Governance: Blockchain technology is revolutionizing how governments operate by facilitating decentralized and efficient systems for tasks like issuing certificates, managing identities, and streamlining tax collection and voting—all while ensuring the validity and security of the processes (Swan, 2015).
- Internet of things (IoT) and 5G: Although IoT systems have many advantages due to the use of centralized architecture, there are many issues, such as single point of failure and security. The transition from IoT to blockchain technology has numerous benefits. Therefore a lot of researches has been proposed on combination of blockchain and IoT (Panarello et al., 2018) and so far, several articles and reviews on the blockchain with IoT (Lee & Lee, 2017; Zhang & Wen, 2017), smart city (Xie et al., 2019), smart home (Moniruzzaman et al., 2020), Internet of vehicle (Mendiboure et al., 2020), and industrial IoT (Alladi et al., 2019) have been presented. Research is also exploring how blockchain can enhance the speed of communication in IoT networks, while simultaneously investigating its role in automating industrial processes that leverage 5G-enabled IoT devices (Mistry et al., 2020), and studies have also been done on blockchain integration with 5G networks (Nguyen et al., 2020; Tahir et al., 2020).
- Privacy and security: Blockchain can be used to increase security and scalability in the field of big data (Puthal et al., 2018). While ensuring transactional privacy remains a challenge for blockchain technology, there are proposed methods to mitigate this issue (Moser, n.d.).
- Other applications: Other blockchain-based applications are not listed above, such as applications in education (Bdiwi et al., 2017; Sharples & Domingue, 2016), business, and industries (Hardjono & Smith, 2016; Tapscott & Tapscott, 2017).

1.4 Security services and blockchain-based solutions

Today, due to the expansion of distributed systems the provision of security services has become very important. Security services are operated by centralized controllers. This centralization confronts several security challenges. Blockchain is a technology that can solve such challenges. In this section, we explain a brief definition of security services, and then we will describe the techniques that provide this type of service. In the concluding part of this section, we will discuss some of their difficulties and shortcomings. We show how blockchain can address these challenges by giving examples of the presented blockchain-based methods.

1.4.1 Authentication and confidentiality services

Authentication verifies the legitimacy of a user's identity, ensuring they are who they say they are. Confidentiality, on the other hand, safeguards the privacy of message content, restricting access only to authorized users. Confidentiality and authentication can be presented using public key cryptography (Salman et al., 2018). In the following, we will consider the mechanisms for providing these security services:

Using public key encryption (asymmetric encryption) is one of the ways to provide security services. This section explores a method for secure communication that relies on two types of keys: public keys, which are widely shared, and private keys, which are kept secret. Public key infrastructure (PKI) acts as a system for managing these keys and ensuring their trustworthiness. We will delve into the two main approaches within PKI: certificate authorities (CAs) (Cooper et al., 2008) and the web of trust (WoT) (Wikipedia, n.d.). CAs are centralized, while WoT is decentralized. While CA-based PKI offers a convenient approach, it faces limitations, such as dependence on a single trusted entity (creating a single point of failure), potential costs associated with certificates, and limitations in transparency for users regarding key management. In a WoT system, new users can only gain access by obtaining a recommendation from an existing, trusted member of the network. However, both methods have limitations, such as the inability to identify retention, which means that it is possible to impersonate the identity or public key of an already registered user (Ellison & Schneier, 2000).

The suitability of blockchain technology for PKI is due to its features, such as distribution and non-reproducibility. These features have solved the problems of traditional PKI-based methods. Some methods are presented for achieving blockchain-based PKI, such as the method presented in Matsumoto and Reischuk (2017), which uses the CA behavior in the blockchain database to facilitate the tracking and monitoring of CA for blockchain users and reduce the problem of trust in the traditional CA-based algorithm, and the proposed method in Gan (2017), which provides a key-based authentication system that uses a blockchain that is private to save information such as the latest public node keys. The distributed PKI (DPKI) method (Allen et al., 2015) focuses on the blockchains as a distributed and unreliable database. This method eliminates the need for CA.

Identity-based encryption (IBE) is an encryption scheme, and the public key in this method is the node ID. The identifier can be the name of a node or any desired string (Boneh & Franklin, 2001). Hierarchical identity-based encryption (HIBE) employs a unique public key structure where user identities are arranged in a hierarchy (Lewko & Waters, 2011). A major drawback of these methods is their reliance on a private key generator (PKG) to create the master key. This centralized approach introduces a single point of failure and creates a trust dependency on the PKG, similar to the challenges faced with CAs in PKI.

Blockchain technology offers a potential solution to the limitations of HIBE by leveraging its decentralized nature. By eliminating the need for a central PKG, blockchain empowers each user to generate their own keys. This distributed approach eliminates the single point of failure inherent in PKG-based HIBE and fosters a trustless system, removing the dependency on a single trusted entity (Fotiou & Polyzos, 2016). Proposed is blockchain-based IBE which provides distributed security for information-centric networking (ICN) networks.

1.4.2 Privacy services

Privacy empowers users to control the disclosure of their data, preventing unauthorized access and unwanted processing. Data access management establishes guidelines that determine who can access specific data, under what conditions, and for what timeframe. Access control lists (ACLs) are a widely used approach to data access management,

allowing granular control over user permissions. ACLs define who can access what information, under what circumstances, and for how long. In the following, we will point out the proposed mechanisms for providing privacy services:

Using encryption is a conventional method to protect data privacy. One of the encryption methods is the homomorphic method, which produces an encrypted output by using specific types of calculations in the encrypted text (Byun et al., 2007). Hiding the user's identity is another feature of privacy. Li et al. (2006) have presented a structure that provides more powerful privacy guarantees. Facing problems like scalability is one of the challenges of using traditional methods like complex cryptography.

The proposed method in Ouaddah et al. (2017) provides a structure using the immutability and internal autonomy features of blockchain. Outchakoucht et al. (2017) illustrated FiarAccess as a framework that combines the access control model and cryptocurrency blockchain mechanism.

1.4.3 Integrity assurance services

Integrity is related to data protection from unauthorized modifications (Kreuter et al., 2013). Data integrity generally can be obtained using cryptographic techniques and replication. Data encryption makes the data immutable by unauthorized persons and any changes in the data can be detected. However, the problem with this method is that if a person acquires an unauthorized secret key, the attack becomes undetectable. If using the replication technique, the attack to change the data needs to change all the copies that are randomly distributed in several nodes. The inability to track intruders and single point of failure are the challenges of these methods that can be solved by blockchain-based methods (Salman et al., 2018).

Blockchain has embedded integrity checks since the sender and miners sign and verify the transactions, respectively. Liu et al. (2017) presented the scheme that provides data integrity service and is proper for data owners and data consumers to confirm particular data without needing a third party.

1.4.4 Provenance services

Data provenance means where the data have come from and how a special piece of data have been generated. This service is used in various fields, such as healthcare and legal proceedings (Asghar et al., 2011), and allows efficient tracking of data with their ownership and uses over the network.

There are traditional methods (Asghar et al., 2011; Muniswamy-Reddy et al., 2006; Suen et al., 2013) for providing data provenance. For example, Suen et al. (2013) proposed a mechanism called S2Logger which captures, visualizes and analyzes data events. This mechanism enables tracking of the data in distributed environments. Asghar et al. (2011) presented a secure data provenance layout that uses a type of search to provide confidentiality of data origination.

Features of blockchain technology, like recording events in the system permanently, enable techniques based on this technology to present data origination services.

BlockCloud (Tosh et al., 2019) is a blockchain-based framework for cloud computing that retains the history of data operation. Javaid et al. (2018) proposed BlockPro approach which uses unique hardware fingerprints to make data provenance and blockchain to tolerate data tampering attacks.

1.5 Blockchain challenges and security threats

While blockchain technology offers innovative solutions to security challenges in various applications, it also faces hurdles that limit its widespread adoption (Bhushan et al., 2021; Salman et al., 2018). To fully realize its potential in the security domain, these challenges must be addressed. In this context, it is important to explore these challenges in more detail, as described in Dasgupta et al. (2019) and Gugueoth et al. (2023). Additionally, we must also consider security threats in the blockchain. There are several categories to categorize security attacks (Bhushan et al., 2021; Dasgupta et al., 2019). In the following, we point to some of the security threats that are discussed in Dasgupta et al. (2019).

1.5.1 Blockchain challenges

- Scalability: scalability remains a significant hurdle for blockchain networks. The growing number of users, demanding security services, and inherent system dynamics all contribute to this challenge. Ongoing research is crucial to identify solutions that can effectively scale blockchain technology.
- User interaction: Users interact with the blockchain-based application in a complex manner; this has become a challenge for nontechnical users and requires user interface improvements.
- Computations and Mining Nodes: Both the extraction process and the cryptography (decryption and signature) used in blockchain technology require high computations. In this regard, resource allocation methods in the extraction process are presented. In addition, simple encryption methods can be proposed to reduce the computations required for encryption.
- Energy consumption: High energy consumption by consensus protocols in blockchain networks threatens the stability of networks.
- Time Consumption: Mining and consensus are time-consuming processes in blockchain. The methods have been proposed that do not require distributed consensus by making decisions locally. Also, the use of cryptography and security techniques increases the complexity of time; thus to reduce the time complexity needed for encrypting, the enhancement of faster extraction and processing techniques needs to be investigated further.
- Communication Overhead: Communication overhead and storage space are other challenges in using blockchain security. Program dynamics often change the source of data and ACLs, in addition, transactions and blocks that are frequently distributed over the network add to the communication overhead on the network.

- Security Issues: Blockchain technology is widely accepted as a means to improve end applications. However, it still remains vulnerable to potential attacks. But in most cases, transactions link the user's identity to the items, such as the public key, providing some level of anonymity. Providing a fully anonymous method remains a challenge that requires further research.
- Upgrade and Fork Management: Managing network forks and coordinating upgrades while maintaining consensus among participants can be complex.

1.5.2 Blockchain security threats

- Vulnerability of cryptographic operation: Symmetric key cryptography and cryptographic hashing are two cryptography methods that are used in blockchain implementations, thus cryptographic key and hashing function vulnerabilities both are blockchain challenges.
- Identify vulnerability: The attacks that compromise the identity of the blockchain user are known as identity.
- Manipulation-based attacks: The attacks that separate a node or nodes from the blockchain network for hostile purposes.
- Reputation-based attack: Changing the validity of a blockchain user from negative to positive is another challenge related to the blockchain network that leads to framework deception.
- Quantum vulnerability: Quantum computing threatens cryptographic algorithms, and since the blockchain uses these algorithms, it is vulnerable.
- Malware attacks: Another threat to the blockchain is malware that tries to disable the blockchain or steal digital currency. The malware, once added to the blockchain, may become impossible to remove due to blockchain immutability.
- Application vulnerability: The integration of third-party applications with blockchain networks introduces potential security vulnerabilities. Among these attacks, we can mention use case design flaws, coding errors, past attacks, and countermeasures.

1.6 Conclusion

Today, control and management in different networks without a central authority have many advantages. With the advent of blockchain technology, it is envisaged to replace centralized systems soon. Due to the extensive use of blockchain in different applications, communication between different blockchains will be necessary. The nature of this technology is such that it can improve security in different applications. This chapter explores the fundamentals of blockchain technology, delving into its core concepts, security challenges, and potential threats. We will also examine cross-blockchain interactions, exploring how different blockchain networks can communicate and collaborate.

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